

$$1. a) C_1 = \alpha W \log_2 \left(1 + \frac{P_1}{\alpha W N_0} \right)$$

$$C_2 = (1-\alpha) W \log_2 \left(1 + \frac{P_2}{(1-\alpha) W N_0} \right)$$

$$C = C_1 + C_2 = \alpha W \log_2 \left(1 + \frac{P_1}{\alpha W N_0} \right) + (1-\alpha) W \log_2 \left(1 + \frac{P_2}{(1-\alpha) W N_0} \right)$$

$$b) \frac{dC}{d\alpha} = \frac{W}{\ln 2} \left[\ln \left(1 + \frac{P_1}{\alpha W N_0} \right) + \alpha \left(\frac{1}{\alpha + \frac{P_1}{W N_0}} - \frac{1}{\alpha} \right) - \left(\ln \left(1 + \frac{P_2}{(1-\alpha) W N_0} \right) + (1-\alpha) \left(\frac{1}{(1-\alpha) + \frac{P_2}{W N_0}} - \frac{1}{1-\alpha} \right) \right) \right]$$

$$= \frac{W}{\ln 2} \left[\left(\ln \left(1 + \frac{P_1}{\alpha W N_0} \right) + \left(1 + \frac{P_1}{\alpha W N_0} \right)^{-1} \right) - \left(\ln \left(1 + \frac{P_2}{(1-\alpha) W N_0} \right) + \left(1 + \frac{P_2}{(1-\alpha) W N_0} \right)^{-1} \right) \right] = 0$$

$$\text{Let } f(x) = \ln(1+x) + (1+x)^{-1}$$

$$\frac{df}{dx} = \frac{1}{1+x} - \frac{1}{(1+x)^2} = \frac{x}{(1+x)^2}$$

when $x > 0$, $\frac{df}{dx}$ must be greater than 0

$\Rightarrow f$ is monotonously increasing following increasing x

$$\Rightarrow \frac{P_1}{\alpha} = \frac{P_2}{1-\alpha} \Rightarrow \frac{\alpha}{1-\alpha} = \frac{P_1}{P_2}$$

$$2. a) T_{fr} = \frac{100 \text{ bits/min}}{2400 \text{ bits/s}} = \frac{1}{1440}$$

$$\text{maximum terminal} = \frac{1}{2T_{fr}} = 720$$

$$b) \text{maximum terminal} = \frac{1}{T_{fr}} = 1440$$

$$3. a) T_{fr} = \frac{b}{R} = \frac{125 \text{ KB}}{100 \text{ Mbps}} = 10 \text{ ms}$$

$$\rho = G e^{-G}$$

when ρ gets maximum, $G = 1 = \lambda \tau T_{fr}$

$$\Rightarrow \lambda \tau = \frac{1}{T_{fr}} = 100 \text{ bits/s}$$

$$b) \frac{1}{2e} = G e^{-G}$$

$$G \approx 0.225 \text{ or } 2.72$$

$\Downarrow$

$$\lambda \tau = 22.5 \text{ or } 272 \text{ bits/s}$$

4. a) decreasing b) decreasing

c) increased to 1: increasing
decreased to $\frac{1}{4}$: decreasing

5. Avoid collision: interframe space, contention window, and ACK

CSMA/CA: station doesn't have any means to check the successful arrival of data but CSMA/CD need.

step	S	D(z), P(z)	D(y), P(y)	D(v), P(v)	D(w), P(w)	D(u), P(u)	D(t), P(t)
0	x	8, x	6, x	3, x	6, x	∞	∞
1	x, v	8, x	6, x	3, x	6, x	5, v	7, v
2	x, v, u	8, x	6, x	3, x	6, x	5, v	6, u
3	x, v, u, y	8, x	6, x	3, x	6, x	5, v	6, u
4	x, v, u, y, w	8, x	6, x	3, x	6, x	5, v	6, u
5	x, v, u, y, w, t	8, x	6, x	3, x	6, x	5, v	6, u
6	x, v, u, y, w, t, z	8, x	6, x	3, x	6, x	5, v	6, u

	x	y	z		x	y	z		x	y	z
x	0	2	7	x	0	2	3	x	0	2	3
y	∞	∞	∞	y	2	0	1	y	2	0	1
z	∞	∞	∞	z	7	1	0	z	3	1	0

	x	y	z		x	y	z		x	y	z
x	∞	∞	∞	x	0	2	7	x	0	2	3
y	2	0	1	y	2	0	1	y	2	0	1
z	∞	∞	∞	z	7	1	0	z	3	1	0

	x	y	z		x	y	z		x	y	z
x	∞	∞	∞	x	0	2	7	x	0	2	3
y	∞	∞	∞	y	2	0	1	y	2	0	1
z	7	1	0	z	3	1	0	z	3	1	0